

No.	Item	Unit	Technical Requirements	Particulars and Guarantees
	d) continuous	rms kV	-	
	Energy capability	kJ/kV	(IEC 60099-4, Clause 7.5.5)	
	Total height of arrester	mm	-	
	Total weight of arrester	kg	-	
	Surge counter		Analogue Type complete with leakage current indication	
INSULATOR FOR 36kV SURGE ARRESTER				
	Insulator Strength	kN	-	
	Minimum creepage distance		-	
	i) Specified	mm	-	
	ii) Guaranteed	mm	1116	
	Protected creepage distance	mm	-	
	Housing material		Porcelain	
	Color		Porcelain (Brown)	

G-3.12 72.5 KV OUTDOOR SURGE ARRESTORS (NOT APPLICABLE)**G-3.13 CAPACITIVE VOLTAGE TRANSFORMERS [NOT APPLICABLE]****G-3.13.1 300 KV OUTDOOR CAPACITIVE VOLTAGE TRANSFORMERS (NOT APPLICABLE)****G-3.13.2 145 KV OUTDOOR CAPACITIVE VOLTAGE TRANSFORMER (VT) (NOT APPLICABLE)****G-3.14 CURRENT TRANSFORMERS [NOT APPLICABLE]****G-3.14.1 300 KV OUTDOOR CURRENT TRANSFORMERS (CT) (NOT APPLICABLE)****G-3.14.2 145 KV OUTDOOR CURRENT TRANSFORMER (CT) (NOT APPLICABLE)**

G-3.15 NEUTRAL EARTHING EQUIPMENT**G-3.15.1 NEUTRAL CURRENT TRANSFORMERS FOR STAR WINDING OF POWER TRANSFORMER**

No	Item	Unit	Technical Requirements	Particulars and Guarantees
	Rated Voltage	kV	≥52	
	Manufacturer Name		Refer to E2.16.5	
	Country of Origin			
	IEC standard		IEC61869-1 & 61869-2	
	Rated Primary Current	A	600	
	Rated Frequency	Hz	50	
	Rated short time current	kA/1s	50	
	Peak withstand current	kA	125	
	Power Frequency Withstand Voltage	kV	95	
	Lightning Impulse Withstand Voltage	kV	250	
	Accuracy Class		PX	
	Principal insulating materials		Cast Resin	
	Type of construction		Post Type	

G-3.15.2 NEUTRAL CURRENT TRANSFORMERS FOR NEUTRAL EARTHING EQUIPMENT

No	Item	Unit	Technical Requirements	Particulars and Guarantees
	Rated Voltage	kV	≥36	
	Manufacturer Name		Refer to E2.16.5	
	Country of Origin			
	IEC standard		IEC61869-1 & 61869-2	
	Rated Primary Current	A	2400	
	Rated Frequency	Hz	50	
	Rated short time current	kA/3s	31.5/3	
	Peak withstand current	kA	78.75	
	Power Frequency Withstand Voltage	kV	70	
	Lightning Impulse Withstand Voltage	kV	170	
	Accuracy Class		PX and 5P20	
	Principal insulating materials		Cast Resin	
	Type of construction		Post Type	

G-3.15.3 NEUTRAL ISOLATORS

No	Item	Unit	Technical Requirements	Particulars and Guarantees
1	System Voltage	kV	33	
2	Manufacturer and country of origin		AEM Australia/Pes-tech Malaysia or equivalent	
3	Type		-	
4	Mechanical Endurance Class		M2 – 10,000	
5	Number of breaks per pole		2	
6	Type of contacts		-	
7	Material of contact surface		-	
8	Normal rated current	A	800	
9	Rated short time current	rms kA	40 (3s)	
10	Air gap between poles of one phase	mm	-	
11	Type of operating mechanism		-	
12	Total weight of three-phase isolator complete	kg	-	
13	Charging current breaking capacity	A	-	
14	Magnetising current breaking capacity	A	-	
15	Motor Operating Mechanisms:			
	i) Power consumption kW at 110 V DC	W	-	
	ii) Operating time	s	-	
16	Lightning impulse withstand voltage			
	i) to earth	kV	170	
	ii) across isolating distance	kV	195	

No	Item	Unit	Technical Requirements	Particulars and Guarantees
17	Rated power frequency withstand voltage	kV	70	
18	Creepage	mm	1116	
19	Radio interference voltage	V	-	
20	IP Rating		IP54	
21	Bushings:			
	Manufacturer		-	
	Material		-	
	Minimum clearance in air:			
	a) Phase to earth	mm	≥ 500	
	Nominal design voltage	kV	36	
	Minimum wet flashover for external		170	
	1 min. dry test	kV	70	
22	Terminal Load (Fx-Horizontal, Fy-Vertical, Fz-Transverse)	kN	≥ 2.6 or in agreement with approved Short Circuit Force calculation	

G-3.15.4 NEUTRAL EARTHING RESISTORS

No	Item	Unit	Technical Requirements	Particulars and Guarantees
1	System voltage	kV	33	
2	Manufacturer and country of origin		Refer to E2.16.5	
3	Applied Standard		IEEE 32	
4	Dimensions		-	
5	Overall height	mm	-	
6	Diameter	mm	-	
7	Plate thickness	mm	-	
8	Weight of resistance	kg	-	
9	Frequency	Hz	50	
10	Resistance at 30°C	Ohms	24	24
11	Current rating – 10 sec for 45°C rise	A	800	800
12	Resistance after rated time	Ohms	24	24
13	Insulation level	kV	-	
14	Power Frequency Withstand Voltage for 1 minute	kV	-	
15	Impulse Withstand Voltage (BIL)	kV	-	
16	Max Temperature rise after rated time	°C	-	
17	Resistances Elements			
a	Resistor Alloy	-	-	
18	Enclosure protection		Hot dip galvanized enclosure	

No .	Item	Unit	Technical Requirements	Particulars and Guarantees
	Power Transformer LV	kN	Min. 3	
	Power Transformer TV	kN	Min. 3	
	Shunt reactor HV	kN	N/A	
	Shunt reactor LV	kN	N/A	
Remarks: For tendering purposes only. Design calculation to be submitted during engineering stage for Engineer approval.				

G-3.37 OVERHEAD CRANES IN GIS ROOM [NOT APPLICABLE]**G-3.38 TRANSFORMERS****G-3.38.1 POWER TRANSFORMER**

	Description	Unit	Technical Requirements	Particulars and Guarantees
1	Manufacturer and country of origin		Refer to E2.16.5	
2.1	Continuous Rating			
	i) ONAN	MVA	80	
	ii) ONAF	MVA	120	
2.2	Tertiary Winding	MVA		
2.3	Voltage ratio at no-load HV/LV/TV	kV	132/33	
3.1	Tapping Range		+10%	
3.2	Size of tapping step		1.25%	
3.3	No of Steps		17	
4	Vector group		YNd1	
5	Lightning impulse withstand voltage:			
	i) 275 kV side	Peak kV		
	ii) 132kV side	Peak kV	650	
	iii) 33 kV side	Peak kV	170	
	iv) neutral side	Peak kV	250	
6	Power frequency withstand voltage:			
	i) 275 kV side	rms kV		
	ii) 132 kV side	rms kV	275	
	iii) 33 kV side	rms kV	70	
	iv) neutral side	rms kV	95	
7	On-load voltage control equipment:			
	(a) Type			
	(b) HV or LV winding		HV	
	(c) Power frequency withstand test voltage between:			

	Description	Unit	Technical Requirements	Particulars and Guarantees
	i) first and last contacts of the selector switch	kV	-	
	ii) between any two adjacent contacts of the selector	kV	-	
	iii) between open diverter switch contacts	kV	-	
9	Heating due to sunrays on the basis of 1kW/m ² exposure to sunrays (tank cover plus one side)	kW	-	
10	Winding temperature rise at MCR under service conditions specified	°C	-	
11	Guaranteed maximum hot spot temperature rise in core at rated power under service conditions	°C	-	
12	Maximum hot spot temperature rise in copper at rated power under service conditions	°C	-	
13	Top oil temperature under service conditions specified (a) MCR (b) ONAN rating (c) ONAF/ODAF	°C °C °C	- - -	
14	Temperature gradient at rating specified	°C	-	
15	Flux density in iron at normal voltage and frequency and at normal ratio – (no load) (a) Cores (b) Yokes (c) Volts/turn	Tesla Tesla V	- - -	
16	Magnetising current (approx) at nominal ratio and (a) At 0.9 x nominal voltage (b) At 1.0 x nominal voltage (c) At 1.1 x nominal voltage (d) At 1.2 x nominal voltage	% % % %	- - - -	
17	Harmonic component of magnetizing current at rated voltage as percentage of fundamental (a) Fifth harmonic (b) Seven harmonic	% %	- -	

	Description	Unit	Technical Requirements	Particulars and Guarantees
18	No-load loss (excluding cooling plant losses) at rated voltage, ratio and frequency	kW	-	
19	Copper loss at 75°C and normal ratio and normal ratio and – (a) MCR (b) ONAN rating (c) ONAF/ODAF	kW kW kW	- - -	
20	Auxiliary loss at MCR (Total power input to oil pumps and fans for operation of total installed cooling capacity)	kW	-	
21	Total loss at 75°C and nominal ratio- (a) MCR including input to cooling plant (b) ONAN rating (c) ONAF/ODAF	kW kW kW	- - -	
22	PEAK EFFICIENCY INDEX (PEI) IEC 60076-20	%		
23	Regulation at 75°C and normal ratio (a) At unity power factor (b) At 0.8 lagging power factor	% %	- -	
24	Impedance voltage at 75°C and MCR between HV & LV windings (a) i) at extreme minus tap ii) normal tap iii) extreme plus tap	% % %	- - -	
	(b) Maximum Impedance over tapping range corresponding tap	%	-	
	(c) Minimum Impedance over tapping range corresponding tap	%	-	
25	Equivalent circuit zero sequence impedance: (a) HV-LV Open circuit (b) HV-LV Short circuit (c) LV-TV (d) HV-TV	% % % %	- - - -	
26	Maximum current density in windings at M.C.R (a) HV winding (b) LV winding	A/mm ²	- - -	

	Description	Unit	Technical Requirements	Particulars and Guarantees
	(c) Tapping winding (d) TV winding	A/mm ² A/mm ² A/mm ²	-	
27.1	Maximum current in tertiary winding	A	-	
27.2	Maximum continuous rating of tertiary winding	MVA	-	
28	Noise level (Sound Power)	dBA	Refer to Vol 2- Part III, Section 4 – Contract Appendix E- E5.22- Clause E5.22.5.5- item (a)	
Details of Construction				
29	Types of transformers (shell or core)		-	
30	Core steel (a) Type (b) Grade		- -	
31	Type of winding (a) HV (b) LV (c) TV		- - -	
32	Insulation of (a) HV windings (b) LV windings (c) TV windings		Thermally Upgraded Paper (120°C) Thermally Upgraded Paper (120°C) Thermally Upgraded Paper (120°C)	Thermally Upgraded Paper (120°C) Thermally Upgraded Paper (120°C) Thermally Upgraded Paper (120°C)
33	Insulation of tapping connections		-	
34	Insulation of (a) Yoke bolts (b) Side plates (c) Core plate		- - -	

	Description	Unit	Technical Requirements	Particulars and Guarantees
35	Safe working temperature of core plate insulation	°C	-	
36	Winding connections brazed or crimped (see Volume 2 - Part III, Section 4 – Contract Appendix E [<i>Employer's Technical Requirements</i>], 5.22.)		-	
37	For fault with zero impedance external to the transformer and infinite system (a) Maximum axial force (b) Maximum tensile stress in winding	kN N/m ²	- -	
38	Is facility provided for adjustment of axial pressure on windings	Yes/No	-	
39	Thickness of transformer tank (see Volume 2 - Part III, Section 4 – Contract Appendix E [<i>Employer's Technical Requirements</i>], 5.22) (a) Sides (b) Bottom	mm mm	- -	
40	Material used for gaskets for oil tight joints		Viton (green color) or equivalent	
41	Make of gaskets for oil tight joints		-	
42	Vacuum to which the tank and radiators can withstand	kPa	-	
43	Positive pressure that tank and radiators can withstand	kPa	-	
Radiators, Valves, Fans and Pumps				
44	Thickness of radiator plates and/or cooling tubes:	mm	-	
45	Valves type/material: (a) 100mm and below (b) Above 100mm		- -	
46	Equipment for ODAF cooling state (a) or (b) (a) Radiators on main tank (b) Separate cooler banks		- -	
47	Number of coolers/radiators or cooler banks per transformer		-	
48	Number of air blowers per transformer		-	
50	Speed of air blowers	rpm	-	

	Description	Unit	Technical Requirements	Particulars and Guarantees
51	Rating of each air blower motor	kW	-	
52	Starting current of each air blower motor	A	-	
55	Maximum pressure of oil at the inlet to cooler under service conditions	kN/m ²	-	
Oil Pumps				
56	Oil pumps manufacturer		-	
	a. manufacturer		-	
	b. country of manufacture		-	
57	Type and number of oil pumps per transformer			
58	Rated normal output of each oil pump (a) Quantity (b) Head approx.	lit/min metres of oil	- -	
59	Speed of oil pump motor	rpm	-	
60	Continuous rating of oil pump motor	kW	-	
61	Starting current of oil pump motor	A	-	
Oil Volumes and weights				
65	Total oil required including cooler system	litres	-	
66	Volume of oil to fill transformer to above the top yoke	litres	-	
67	Total volume of conservator	litres	-	
68	Volume of oil in conservator between highest and lowest visible	litres	-	
69.1	Weight of active copper	tonnes	-	
69.2	Weight of active iron in core	tonnes	-	
69.3	Weight of core and winding assembly	tonnes	-	
70	Weight of each cooler bank complete with oil if mounted separately from transformer	tonnes	-	
71	Total weights of complete transformers, including attached coolers, voltage regulating equipment, all fittings and oil	tonnes	-	

	Description	Unit	Technical Requirements	Particulars and Guarantees
72	Weight of transformer arranged for transport	tonnes	-	
76	Minimum operating life		Min. 40 years	Min. 40 years
Accessories				
	Temperature Indicator (Oil & Winding) Manufacturer		AKM Sweden Messko Germany or equivalent	
	Type and make of air blower		Ziehl Abegg Germany/ California Turbo USA/ Krenz-Venz Germany	
	Main Tank Pressure relief device Manufacturer		Qualitrol USA/ Messko Germany or equivalent	
	OLTC Pressure relief device Manufacturer		Qualitrol USA/ Messko Germany or equivalent	
	Oil level indicator manufacturer		Qualitrol USA/ Messko Germany or equivalent	
	Buchholz relay manufacturer		EMB Germany/ ETI Italy / Comem Italy	
	Conservator airbag manufacturer		Atmoseal France/ Musthane France or equivalent	
	Manufacture, type and class of oil to IEC 60296 (see Clause E5.22.5.15.1 - <i>Transformer Oil – Part III: Section 4 – Contract Appendix E5.22 Power Transformers and Ancillaries</i>)		Nynas Nytro Gemini X, Sweden / Shell Diala S4 ZX-1, Hong Kong / Ergon Hyvolt III, USA or equivalent	
	Main tank and OLTC Conservator air bag conventional breather (Preferred color of silica gel: Orange)		-	

	Description	Unit	Technical Requirements	Particulars and Guarantees
	On-line winding temperature monitoring system (Fibre Optic)		Qualitrol Neoptix T/Guard 408XT or equivalent	
Transformer Parts Subject to Short-Circuit Test				
77	Brief description of transformer or parts thereof subjected to short-circuit test or for which short-circuit calculations are available.		-	

G-3.38.1.1 ON LOAD TAP CHANGER

	Item	Unit	Technical Requirements	Particulars and Guarantees
1	Type / Manufacturer of the vacuum tap changer		MR Germany / ABB Sweden or equivalent	
2	Rated through-current: i) relevant rated step voltage	A	-	
3	Maximum rated through-current: i) relevant rated step voltage	A	-	
4	Maximum admissible short-circuit current for 3s		-	
5	Type of transition impedance (resistor or reactor)		-	
6	Type and material of contacts		-	
7	Guaranteed life of contacts (operations)		-	
8	Gas actuated relays manufacturer		-	

G-3.38.1.2 TRANSFORMER BUSHING

G-3.38.1.2.1 500 KV TRANSFORMER BUSHING (NOT APPLICABLE)

G-3.38.1.2.2 275 KV TRANSFORMER BUSHING [NOT APPLICABLE]

G-3.38.1.2.3 132 KV TRANSFORMER BUSHING

	Item	Unit	Technical Requirements	Particulars and Guarantees
1	System Voltage	kV	132	
2	Manufacturer & Maker (model)		ABB Micafil, Switzerland (RTKF) / HSP, Germany (STEFt) / ABB, Sweden (GSA/GSB)/ Trench, China	
3	Bushing type		Solid rod	
4	Maker's type reference		-	
5	Insulator material (solid, oil-paper etc)		RIP	
6	Rated voltage	kV	145	
7	Rated current	A	2000	
8	Manufacturer of porcelain		-	
9	Length of insulator (overall)	mm	-	
10	Weight of insulator	kg	-	
11	Electrostatic capacity of complete bushings	pF	-	
12	Rated short-time current 1s	kA	Min. 50 kA	
13	Power frequency test voltage dry	kV	275	
14	Lightning impulse withstand voltage (1.2/50 microseconds)	kV	650	
15	50 Hz wet withstand voltage without arcing horns	kV	275	
16	Cantilever strength	kN	4	
17	Total creepage distance of outdoor shedding (>31mm/kV)	mm	> 4495	
18	Protected creepage distance of shedding	mm	-	
19	Terminal load (Fx-Horizontal, Fy-Vertical, Fz-Tranverse)	kN	≥ 4	

G-3.38.1.2.4 33 KV TRANSFORMER BUSHING

	Item	Unit	Technical Requirements	Particulars and Guarantees
1	System Voltage	kV	33	
2	Manufacturer & Maker (model)		ABB Micafil, Switzerland (RTKF) / HSP, Germany (STEft) / ABB, Sweden (GSA/GSB)/Trench, China	
3	Bushing type		Solid Rod	
4	Maker's type reference		-	
5	Insulator material (solid,oil-paper etc)		RIP	
6	Rated voltage	kV	52	
7	Rated current	A	2500	
8	Manufacturer of porcelain		-	
9	Length of insulator (overall)	mm	-	
10	Weight of insulator	kg	-	
11	Electrostatic capacity of complete bushings	pF	-	
12	Rated short-time current 3s	kA	Min. 31.5 kA	
13	Power frequency test voltage dry	kV	95	
14	Lightning impulse withstand voltage (1.2/50 microseconds)	kV	250	
15	50 Hz wet withstand voltage without arcing horns	kV	95	
16	Cantilever strength	kN	3	

	Item	Unit	Technical Requirements	Particulars and Guarantees
17	Total creepage distance of outdoor shedding (>31mm/kV)	mm	> 1612	
18	Protected creepage distance of shedding	mm	-	
19	Terminal load (Fx-Horizontal, Fy-Vertical, Fz-Tranverse)	kN	≥ 3	

G-3.38.1.2.5 HV NEUTRAL TRANSFORMER BUSHING

	Item	Unit	Technical Requirements	Particulars and Guarantees
1	System Voltage	kV	Neutral	
2	Manufacturer & Maker (model)		ABB Micafil, Switzerland (RTKF) / HSP, Germany (STEft) / ABB, Sweden (GSA/GSB) / Trench, China	
3	Bushing type		Solid rod	
4	Maker's type reference		-	
5	Insulator material (solid,oil-paper etc)		-	
6	Rated voltage		52	
7	Rated current		2500	
8	Manufacturer of porcelain		-	
9	Length of insulator (overall)	mm	-	
10	Weight of insulator	kg	-	

	Item	Unit	Technical Requirements	Particulars and Guarantees
11	Electrostatic capacity of complete bushings	pF	-	
12	Rated short-time current 3s	kA	-	
13	Power frequency test voltage dry	kV	95	
14	Lightning impulse withstand voltage (1.2/50 microseconds)	kV	250	
15	50 Hz wet withstand voltage without arcing horns	kV	-	
16	Cantilever strength	kN	3	
17	Total creepage distance of outdoor shedding (>31mm/kV)	mm	>1612	
18	Protected creepage distance of shedding	mm	-	
19	Terminal load (Fx-Horizontal, Fy-Vertical, Fz-Tranverse)	kN	≥ 3	

G-3.38.2 275/33 POWER TRANSFORMER (NOT APPLICABLE)

G-3.38.2.1 ON LOAD TAP CHANGER (NOT APPLICABLE)

G-3.38.2.2 TRANSFORMER BUSHING

G-3.38.2.2.1 500 KV TRANSFORMER BUSHING (NOT APPLICABLE)

G-3.38.2.2.2 275 KV TRANSFORMER BUSHING (NOT APPLICABLE)

G-3.38.2.2.3 33 KV TRANSFORMER BUSHING (NOT APPLICABLE)

G-3.38.2.2.4 HV NEUTRAL TRANSFORMER BUSHING (NOT APPLICABLE)

G-3.38.3 AUTO TRANSFORMER (NOT APPLICABLE)

G-3.38.3.1 ON LOAD TAP CHANGER (NOT APPLICABLE)

G-3.38.3.2 TRANSFORMER BUSHING (NOT APPLICABLE)

G-3.38.3.2.1 300 KV TRANSFORMER BUSHING [NOT APPLICABLE]
G-3.38.3.2.2 145 KV TRANSFORMER BUSHING (NOT APPLICABLE)
G-3.38.3.2.3 33 KV TRANSFORMER BUSHING (NOT APPLICABLE)
G-3.38.3.2.4 HV NEUTRAL TRANSFORMER BUSHING (NOT APPLICABLE)
G-3.38.4 EARTHING TRANSFORMER

No	Item	Unit	Technical Requirements	Particulars and Guarantees
1	Manufacturer and country of origin		Tirathai Thailand / QTC Thailand or equivalent	
2	Type		Oil immersed ONAN type	
	i) HV & HVN bushing manufacturer		-	
	ii) LV & LVN bushing manufacturer		-	
3	Rated short time (10 sec.) current ($3I_0$)	A	800	
4	Current density in the interconnected star winding at rated short time (10 sec.) current	A/mm ²	-	
5	Current density in the interconnected star winding at MCR and 5 second fault rating simultaneously	A/mm ²	-	
6	Maximum flux density in core iron at normal voltage and frequency	Tesla	-	
7	MCR of 415V winding	kVA	200	
8	Rated frequency	Hz	50	
9	Rated voltages of windings:			
	i) Nominal (HV/LV)	kV	33/0.415	
	ii) Maximum (HV/LV)	kV	36/0.460	

No	Item	Unit	Technical Requirements	Particulars and Guarantees
10	No load losses at rated voltage & frequency	kW	-	
11	Load loss at 75°C at MCR of lower voltage winding	kW	-	
12	Impedance voltage at MCR of lower voltage winding between HV and LV winding	%	-	
13	Resistance of higher voltage winding at 75°C	Ohms per phase	-	
14	Zero phase sequence impedance at 75°C with LV windings open circuited	Ohms per phase	10 (no tolerance)	
15	Audible sound level (ONAN) (Sound pressure)	dBA	Refer to Vol 2-Part III, Section 4 – Contract Appendix E-E5.22- Clause E5.22.5.5-item (b)	
16	Total oil required	litres	-	
17	Weight of complete transformer (including oil)	tonnes	-	
18	Height from base to bottom of neutral bushing insulator	mm	-	
19	Transformer oil Manufacturer		Nynas Nytro Gemini X, Sweden / Shell Diala S4 ZX-1, Hong Kong / Ergon Hyvolt III, USA	

No	Item	Unit	Technical Requirements	Particulars and Guarantees
20	Oil level indicator Manufacturer		Qualitrol USA / Messko Germany or equivalent	
21	Buchholz relay Manufacturer		EMB Germany / ETI Italy / Comem Italy	
22	Pressure relief device Manufacturer		Qualitrol USA / Messko Germany or equivalent	
23	Conservator air bag breather (remark: orange color silica gel)		-	
24	HV bushing:			
	Cantilever strength	kN	-	
	Terminal load (Fx-Horizontal, Fy-Vertical, Fz-Tranverse)	kN	≥1.0 or in agreement with approved Short Circuit Force calculation	
25	HVN bushing:			
	Cantilever strength	kN	-	
	Terminal load (Fx-Horizontal, Fy-Vertical, Fz-Tranverse)	kN	≥1.0 or in agreement with approved Short Circuit	

Trip coil voltage	V dc				
Is the circuit breaker trip free?	Yes/ No				
Type of arcing contacts or arc control device or method					
Type of main contact surfaces					
Does magnetic effect of load current increase contact pressure?					
Number of breaks per phase					
Voltage distribution between breaks	%				
Length of each break	mm				
Length of stroke	mm				
DIMENSIONS, ETC.					
Weight of whole circuit breaker equipment and all fitting as in service	kg				
Maximum shock load imposed on floor or foundations when opening under fault condition (state whether tension or compression)	kg				
Maximum pressure rise in circuit breaker due to the making or breaking of rated current	kg				
Routine pressure test on circuit breaker tanks or containers	kN/m ²				
Pressure type test on circuit breaker tanks or container	kN/m ²				
Interrupting gas pressure (normal) at 20°C	kN/m ²				
Limit of gas pressure at 20°C					

- Maximum	kN/m ²					
- Minimum	kN/m ²					
Minimum clearances						
- Between phases	mm					
- Phase to earth	mm					
- Between live parts of one phase	mm					
Distance necessary for breakers withdrawal (if applicable)	mm					
Minimum creepages:						
- Phase to earth	mm					
- Phase to phase	mm					
- Across circuit breaker poles	mm					
Materials of moving contact tension rod						
Loading of heaters for circuit breaker						
Period of time equipment has been in commercial operation						
Details of floor plates or rails – if provided						
Overall dimensions of each breaker unit						
33kV Indoor						
- Height	mm					
- Length	mm					
- Width	mm					

G-3.51.2 33kV VOLTAGE TRANSFORMER

No.	Description	Unit	132/33kV BESS Power Transformer	Samajaya 132/33kV Transformer T3	Busbar	BESS and Auxiliary Transformer [Tenderer to specify]
1.	Manufacturer			Ritz, Germany/Ritz, Austria/Stemar, Australia		
2.	Type /model					
3.	Nominal voltage rating		$\frac{33kV}{\sqrt{3}} / \frac{110V}{\sqrt{3}} / \frac{110V}{\sqrt{3}}$	$\frac{33kV}{\sqrt{3}} / \frac{110V}{\sqrt{3}} / \frac{110V}{\sqrt{3}}$	$\frac{33kV}{\sqrt{3}} / \frac{110V}{\sqrt{3}}$	
4.	Rated burden per phase					
	i. Protection	VA				
	ii. Metering	VA				
	iii. Main Tariff Metering	VA				
	iv. Check Tariff Metering	VA				
5.	Accuracy Class					
	i. Protection	%	3P	3P	N/A	
	ii. Metering	%	0.5	N/A	0.5	
	iii. Main Tariff Metering	%	NA	0.2	NA	
		%	NA	0.2	NA	

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	iv. Check Tariff Metering					
6.	Maximum ratio error with rated burden and 5% normal primary voltage					
	i. Protection	%				
	ii. Metering	%				
	iii. Main Tariff Metering	%				
	iv. Check Tariff Metering	%				
7.	Maximum phase angle error with rated burden and 5% normal primary voltage					
	i. Protection	%				
	ii. Metering	%				
	iii. Main Tariff Metering	%				
	iv. Check Tariff Metering	%				
8.	Overvoltage factor					
9.	Overvoltage duration	s				
10.	Total weight of 3 phase units	kg				
11.	Total weight of 1 phase units	kg				

G-3.51.3 33kV CURRENT TRANSFORMER

No	Description	Units	Busbar Zone Protection	Biased Differential	Overcurrent /Earth fault	Restricted Earth Fault	Instruments
	<u>132/33kV BESS Power Transformer Incomer</u>						
1.	Manufacturer						
2.	Type /model						
3.	Ratio		2500/1A	2400/1A	2400/1A	2400/1A	2400/1A
4.	Maximum output	VA			15		15
5.	Knee Point Voltage	V	≥300	≥300		≥300	
6.	Accuracy class		PX	PX	5P20	PX	0.2
7.	Short time thermal current (1s)	kA					
8.	Rated accuracy limit factor						
9.	Voltage withstand level	kV					
10.	Impulse withstand voltage	kV					

No	Description	Units	Busbar Zone Protection	Biased Differential	Overcurrent /Earth fault	Restricted Earth Fault	Tariff metering
	<u>Samajaya 132/33kV Power Transformer T3 Incomer</u>						
1.	Manufacturer		Ritz, Germany/Ritz, Austria/Stemar, Australia	Ritz, Germany/Ritz, Austria/Stemar, Australia	Ritz, Germany/Ritz, Austria/Stemar, Australia	Ritz, Germany/Ritz, Austria/Stemar, Australia	Ritz, Germany/Ritz, Austria/Stemar, Australia
2.	Type /model						
3.	Ratio		2500/1A	2400/1A	2400/1A	2400/1A	2400/1A
4.	Maximum output	VA			15		15
5.	Knee Point Voltage	V	≥300	≥300		≥300	
6.	Accuracy class		PX	PX	5P20	PX	0.2s
7.	Short time thermal current (1s)	kA					
8.	Rated accuracy limit factor						
9.	Voltage withstand level	kV					
10.	Impulse withstand voltage	kV					

No	Description	Units	Busbar Zone Protection	Overcurrent /Earth Fault	Instruments	Others [Tenderer to specify]
	<u>Auxiliary Transformer Feeder</u>					
1.	Manufacturer					
2.	Type /model					
3.	Ratio		2500/1A			
4.	Maximum output	VA				
5.	Knee Point Voltage	V	≥300			
6.	Accuracy class		PX			
7.	Short time thermal current (1s)	kA				
8.	Rated accuracy limit factor					
9.	Voltage withstand level	kV				
10.	Impulse withstand voltage	kV				

Guaranteed Plant Emissions	Unit	Guaranteed Value
Maximum Guaranteed Noise Emissions of BESS at 100% MCR at the 1 meter from equipment	dB	≤85
Maximum Guaranteed Noise Emissions of BESS at 100% MCR at the closest Site Boundary	dB	≤70

3.0. Annual Performance Guarantees

- 3.1. The guaranteed values for the BESS at the beginning of each operation year for the BESS Annual Performance Tests are set out in the table below. The BESS Annual Performance Tests shall be carried out as set forth in Contract Appendix E5B (Electrical Works – Battery Energy Storage System).
- 3.2. BESS Operating Parameters which influence the Guaranteed values are limited to only the ones listed in Table 3.2 below.

Description	Values
Number of Equivalent Cycles/Year	365
Average SOC	20 – 90%
Others: Tenderer to Specify	

Table 3.2: BESS Operating Parameters which influence the Guaranteed values

- 3.3. If BESS Equivalent Cycles in a given year are more than the listed values in Table 3.2, the Guaranteed Energy Capacity will be adjusted and referred to during any annual performance test.
- 3.4. Formula for adjustments based on actual usage of Number of Equivalent Cycles/years are as below:

Adjusted SOH formula for the BESS Plant are as below:

Adjusted SOH formula = [to be defined]

Guaranteed values will not be adjusted with the formula in Section 3.3 if Equivalent Cycles remain the same as the value defined in Section 4.1.

Guaranteed Energy Capacity and Guaranteed Storage Efficiency would be adjusted from the Adjusted SOH as per the formula below:

Adjusted Guaranteed Energy Capacity = Function of Adjusted SOH [to be defined]

Adjusted Guaranteed Storage Efficiency = Function of Adjusted Guaranteed Energy Capacity [to be defined]

- 3.5. The Contractor shall pay to the Employer performance liquidated damages as provided for in Contract Appendix B [Contract Price]:

If at the beginning of each operation year, the Actual Power & Energy Capacity, Actual Storage Efficiency, Actual BESS Availability is below 100% of the respective Guaranteed values and the BESS Auxiliary Consumption – Standby exceeds the respective guaranteed value as specified in Table 3.5a and 3.5b below;

The guaranteed values in Table 3.5a and Table 3.5b shall be calculated based on the BESS operating parameters defined in Table 3.2 above. The guaranteed values shall not be less than the requirements set in Contract Appendix A, section 1.2 [Overview of BESS Plant key technical requirements].

Description	Unit	Year 1 – Year 15
Guaranteed BESS Availability	%	
Guaranteed Import Power	MW	
Guaranteed Export Power	MW	
Guaranteed BESS Auxiliary Consumption – Standby	kW	

Table 3.5a: Guaranteed BESS availability, Import & Export Power and BESS Auxiliary Consumption – Standby

Beginning of Year	Guaranteed Energy Capacity at PCC (MWh)	Guaranteed Storage Efficiency at PCC (%)	Battery Remaining SOH (%)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			

Table 3.5b: Guaranteed Energy Capacity and Guaranteed Storage Efficiency